Convolutional Neural Network for Cardiac Arrhythmia Detection from ECG Signals

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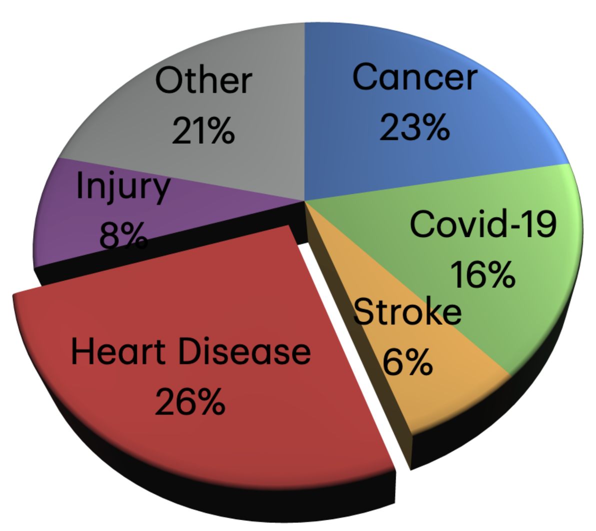
Overview

Cardiac arrhythmia accounts for the largest percentage of premature deaths in the United States. These are heart conditions characterized by uncoordinated electrical pulses in the heart. These pulses can be read through ECG readings and diagnosed by a medical professional. Due to the intermittent and irregular nature of its symptoms, reliable detection is difficult, especially in short or noisy signals. In this work, we investigate machine learning approaches for the automated classification of various types of cardiac arrhythmias. With an emphasis on classifying the type of arrhythmia given a limited signal duration.



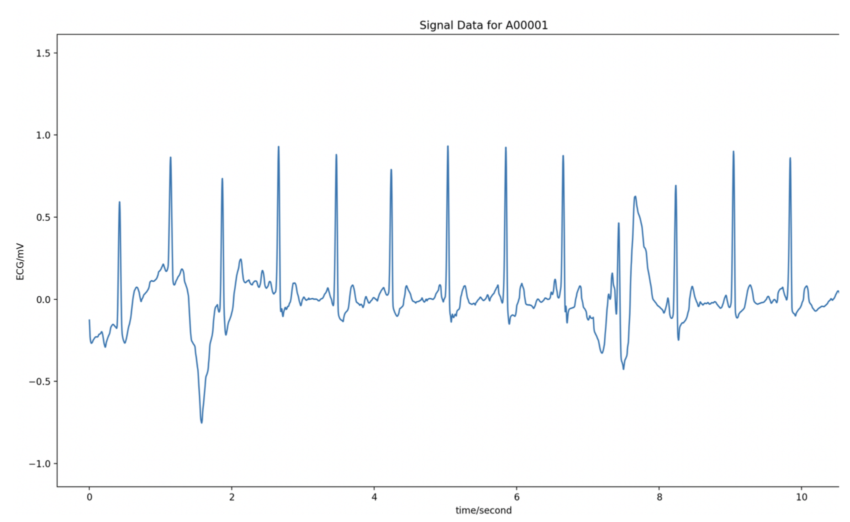


Fig. 1 Leading causes of death in the U.S. **Fig. 3** Sample ECG signal from MIT-BIH database

About the Dataset

The dataset used in this study was sourced from the PhysioNet site and contains an open-source download of a dataset with the following characteristics:

- MIT-BIH Arrhythmia Database
- 12-lead ECGs
- Selected 9 of 63 classes for comparison against baseline paper
- Consists of 45,152 patient ECGs
- ECG segments are 10 sec length
- 500 Hz sampling rate
- Each class has anywhere from 500 to 9000 samples

Preprocessing

Developed a flexible Python preprocessing pipeline to standardize labels and restructure the dataset automatically traversing the dataset tree, operating on paired .hea and .mat files with shared basenames.Used the WFDB library (wfdb.rdheader) to:

- Extract classification metadata
- Identify sampling frequency and lead information
- Removed incomplete or improperly labeled records.

Label Standardization

- Extracted numeric diagnostic codes from header comment lines.
- Cross-referenced codes with the 'SNOMED CT' clinical terminology table.
- Converted numerical identifiers into clinically meaningful arrhythmia labels.
- Reorganized records into class-specific directories (hierarchical dataset structure).

Test/Training Split:

- Applied stratified 80/20 split to preserve class distributions and randomized dataset prior to partitioning.

Classical Model: 1-D Convolutional Neural Network

- 1D Convolutional Neural Networks (CNNs) effectively learn and extract discriminative features directly from raw ECG waveforms.
- Well-suited for time-series data, enabling detection of temporal patterns such as normal sinus rhythm vs. arrhythmic activity.
- Convolutional layers apply learnable filters to capture local waveform features (e.g., QRS complex morphology, rhythm irregularities).
- Pooling layers reduce feature dimensionality, improve computational efficiency, and enhance robustness to small temporal variations.

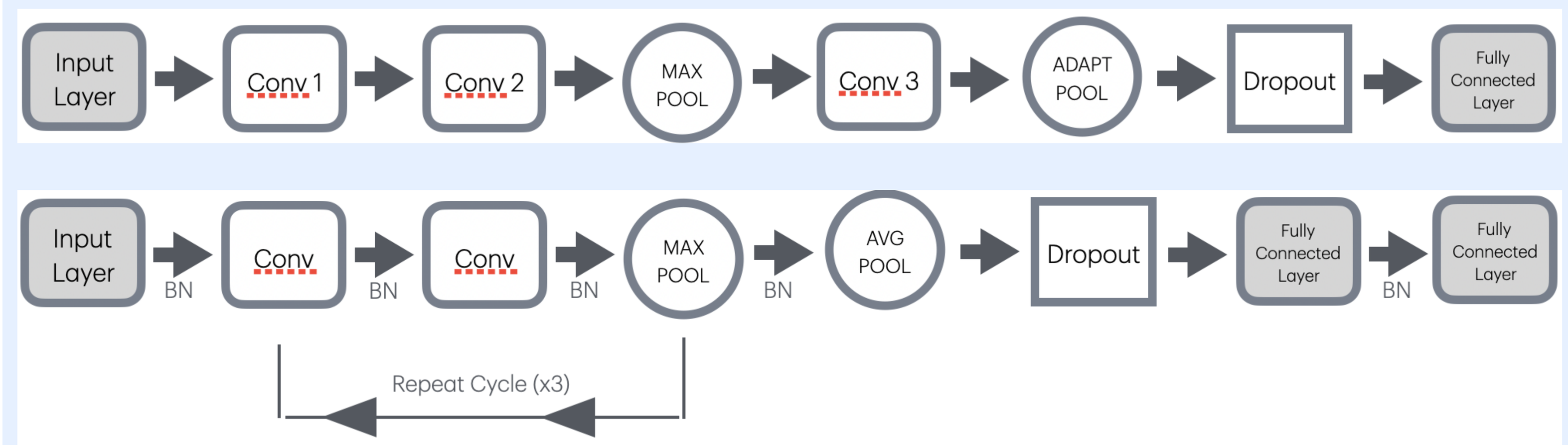


Fig. 2 Baseline Model -1D CNN Architecture

The architecture is modular and includes tunable hyperparameters:

- Number of filters: F
- Kernel size: k
- Dropout rate: p
- Batch size

This flexibility enables systematic performance optimization.

For an input ECG signal $x[n]$, a 1D convolutional layer computes:

$$y_j[n] = \sum_{i=1}^{C} \sum_{m=0}^{k-1} w_{j,i}[m]\, x_i[n-m] + b_j$$

- k = kernel size
- $w_{j,i}$ = learned filter weights
- b_j = bias
- $y_j[n]$ = output feature map

Results

Training both models on several different seeds resulted in the following performance scores:

Table 1: Test Model Performance Across Random Seeds

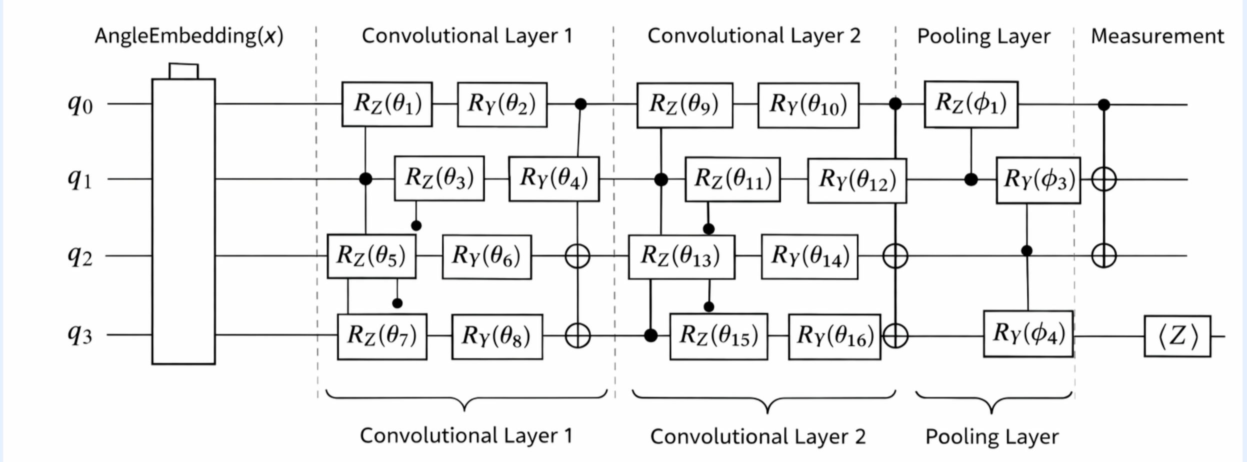
Seed	13	23	33	43	53	Avg
Accuracy	99.77%	99.87%	99.86%	99.87%	99.84%	99.84% ± 0.04%
Precision	99.06%	99.93%	99.85%	99.88%	99.79%	99.70% ± 0.36%
Recall	99.62%	99.83%	99.80%	99.85%	99.80%	99.78% ± 0.09%
F1 Score	99.33%	99.80%	99.82%	99.86%	99.79%	99.72% ± 0.22%

Table 2: Base Model Performance Across Random Seeds

Seed	13	23	33	43	53	Avg
Accuracy	96.31%	99.92%	81.67%	83.13%	89.23%	90.05% ± 7.99%
Precision	70.08%	99.79%	28.40%	58.99%	56.82%	62.82% ± 25.76%
Recall	72.64%	99.74%	33.12%	66.66%	64.08%	67.25% ± 23.77%
F1 Score	70.96%	99.77%	29.66%	52.87%	57.58%	62.17% ± 25.77%

Quantum Convolutional Neural Network

- **Classical Preprocessing** - In this step our goal is to reduce dimensional while preserving diagnostic structure. By decreasing the over 1000 samples from 2 sec long recordings into a feature map with only C number of components.
- **Quantum Feature Encoding** - Each classical feature is encoded into a qubit via parameterized rotations.
- **Quantum Convolutional Layers** - Apply two quibit gates and leverages entanglement to extract hierarchical quantum features and capture correlations between neighboring qubits. These are comparable to spatial features in classical CNNs.
- **Quantum Pooling Layers** - Applies controlled rotations effectively compressing quantum information further.



Quantum Advantages

- **Reduced Dataset Requirements** - Quantum Neural Networks are known to produce effective predictive models with excellent generalization performance even when provided with only a small amount of training data. Leveraging High-dimensional Hilbert spaces and exponentially large state representations allow for this.
- **Resilience to Noise** - QCNNs can have improved performance in dealing with noisy ECG signal samples. Variational quantum circuits can exhibit robustness to certain types of noise due to the fact that they distribute information across amplitudes rather than store them deterministically.

Future Potential Research

While quantum convolutional neural networks have been proven to be somewhat effective in my past research when tasked with binary classification for normal vs Atrial Fibrillation (AF) ECG signals, when tasked with multi-classification my current model is not effective. Through further exploration of quantum circuit configurations and dimensionality reduction, a higher performing quantum model is achievable, however simulation and hardware restrictions make this challenging.

Acknowledgments

[1] Statista. (n.d.). *Deaths from leading causes of death in the United States.* Available: https://www.statista.com/chart/30883/deaths-from-leading-causes-of-death-in-the-united-states/

[2] T. Vu, T. Petty, K. Yakut, M. Usman, W. Xue, F. M. Haas, R. A. Hirsh, and X. Zhao, "Real-time arrhythmia detection using convolutional neural networks," *Frontiers in Big Data*, vol. 6, 1270756, 2023. doi: 10.3389/fdata.2023.1270756

[3] J. Zheng, H. Guo, and H. Chu, "A large scale 12-lead electrocardiogram database for arrhythmia study (version 1.0.0)," *PhysioNet*, 2022. doi: 10.13026/wgex-er52